2.1) $\ell(\mu,\sigma^2) = -\sum \ln x_i - \frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(\ln x_i - \mu)^2$

Say $\beta = \sigma^2$

$$\ell(\mu,\beta) = -\sum \ln x_i - \frac{n}{2}\ln(2\pi\beta) - \frac{1}{2\beta}\sum(\ln x_i - \mu)^2$$

For μ:

$$\frac{d\ell}{d\mu} = -\frac{1}{2\beta}\sum \frac{d}{d\mu}(\ln x_i - \mu)^2$$

$$= -\frac{1}{2\beta}\sum 2(\ln x_i - \mu)(-1)$$

$$\boxed{= \frac{1}{\beta}\sum(\ln x_i - \mu)} = 0$$

$$\sum(\ln x_i - \mu) = 0$$

$$\sum \ln x_i - \sum \mu = 0$$

$$\sum \ln x_i - n\mu = 0$$

$$\hat{\mu} = \frac{\sum \ln x_i}{n}$$

For β:

$$\frac{d\ell}{d\beta} = -\frac{n}{2}\cdot\frac{1}{\beta} + \frac{1}{2\beta^2}\sum(\ln x_i - \mu)^2$$

$$2\beta^2\left(\boxed{= -\frac{n}{2\beta} + \frac{1}{2\beta^2}\sum(\ln x_i - \mu)^2}\right) = 0$$

$$= -n\beta + \sum(\ln x_i - \mu)^2 = 0$$

$$n\beta = \sum(\ln x_i - \mu)^2$$

$$\hat{\beta} = \frac{\sum(\ln x_i - \mu)^2}{n}$$

$\hat{\sigma}^2$